

Test 30: Dirac cone vF scaling vs L

Verdict: PASS | Generated: 2026-09-15 04:40:07 | Suite: AB-Cloud v23 SUPERCOMBO (Julia 1.12.0)

What this test verifies

EXACT-geometry: FIRST DISTINCT Dirac level above the zero tower ~ 1/L (the old $ev[j0+1]-ev[j0]$ hit the valley-degenerate partner ~1e-15 ? gaps 0.0); L/4?? grid up to t31_lmax;

METHOD: EXACT-geometry: FIRST DISTINCT Dirac level above the zero tower ~ 1/L (the old $ev[j0+1]-ev[j0]$ hit the valley-degenerate partner ~1e-15 ? gaps 0.0); L/4?? grid up to t31_lmax;

Test 30 HARDCORE: 11-point fit $R\approx 0.9999$, $v_F=1.9447$, $E\propto L$ cv=0.75%, zero tower 4 ?L.

Result [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS

- Plots
- 'plots/plot_01.png' ? 1600x1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_02.png' ? 1600x1000, ABPlotV23 engine (supersampled)
 - 'plots/plot_03.png' ? 1600x1000, ABPlotV23 engine (supersampled)
 - 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v23.jl --test 30 --no-two-pass

Every computation is logged in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

Independent verification guide

[04:30:03.322] [+16994.351s] build_ab_cloud_hamiltonian: 16x16 (N=256), ≈ 0.5000 , t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:30:04.976] [+16996.005s] build_ab_cloud_hamiltonian: 24x24 (N=576), ≈ 0.5000 , t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:30:06.368] [+16997.396s] build_ab_cloud_hamiltonian: 32x32 (N=1024), ≈ 0.5000 , t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:30:08.104] [+16999.132s] build_ab_cloud_hamiltonian: 40x40 (N=1600), ≈ 0.5000 , t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:30:10.730] [+17001.759s] build_ab_cloud_hamiltonian: 48x48 (N=2304), ≈ 0.5000 , t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:30:15.385] [+17006.413s] build_ab_cloud_hamiltonian: 56x56 (N=3136), ≈ 0.5000 , t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:30:25.362] [+17016.391s] build_ab_cloud_hamiltonian: 64x64 (N=4096), ≈ 0.5000 , t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:30:46.224] [+17037.252s] build_ab_cloud_hamiltonian: 72x72 (N=5184), ≈ 0.5000 , t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:31:31.588] [+17082.616s] build_ab_cloud_hamiltonian: 80x80 (N=6400), ≈ 0.5000 , t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:32:56.543] [+17167.571s] build_ab_cloud_hamiltonian: 88x88 (N=7744), ≈ 0.5000 , t=1.00, W=0.00, torus, 0 vortices, model=:monumental

[04:35:25.008] [+17316.036s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ≈ 0.5000 , t=1.00, W=0.00, torus, 0 vortices, model=:monumental

??

TEST 30 (pass 2, HARDCORE): Dirac cone velocity ? dense-grid audit

??

CONSOLE CAPTURE

11-point fit $R\approx 0.9999$, $v_F=1.9447$, $E\propto L$ cv=0.75%, zero tower check, $E\propto L$ constancy

? RUNTIME: 11 value-only diagonalizations, L up to 96 ? expect ? 8?12 min

L=16: E? = 0.765367 (E?•L = 12.2459), zero_modes = 4, 0.5 s

L=24: E? = 0.517638 (E?•L = 12.4233), zero_modes = 4, 0.5 s

L=32: E? = 0.390181 (E?•L = 12.4858), zero_modes = 4, 0.8 s

L=40: E? = 0.312869 (E?•L = 12.5148), zero_modes = 4, 1.8 s

L=48: E? = 0.261052 (E?•L = 12.5305), zero_modes = 4, 3.9 s

L=56: E? = 0.223929 (E?•L = 12.5400), zero_modes = 4, 9.2 s

L=64: E? = 0.196034 (E?•L = 12.5462), zero_modes = 4, 19.8 s

L=72: E? = 0.174311 (E?•L = 12.5504), zero_modes = 4, 44.4 s

L=80: E? = 0.156918 (E?•L = 12.5535), zero_modes = 4, 83.8 s

L=88: E? = 0.142678 (E?•L = 12.5557), zero_modes = 4, 147.2 s

L=96: E? = 0.130806 (E?•L = 12.5574), zero_modes = 4, 277.7 s

Fit: gap = 12.2188/L + 0.0054, R? = 0.999880, v_F = 1.9447 (?-flux target 2.0)

E?•L: mean = 12.5003, cv = 0.75% ? ? 5%

??

R? > 0.95 (11 pts) PASS ? R? = 0.9999

zero tower == 4 ?L PASS ? |E| ? 1e-8 at every size

E?•L cv ? 5% PASS ? cv = 0.75% (Dirac law: E?•L ? 4? = const)

Test 30b [HARDCORE pass 2]: 3 sub-checks, 0 failed ? PASS